

TrES-3b

R-1435

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_190527 · created 2026-08-01T07:52:35+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2458630.8633 +/- 0.0013 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.193 +/- 0.021
Transit depth (Rp/Rs)^2	3.72 +/- 0.8 [%]
Orbital Inclination (inc)	81.67 +/- 0.55
Airmass coefficient 1 (a1)	1.316 +/- 0.014
Airmass coefficient 2 (a2)	-0.264 +/- 0.069
Scatter in the residuals of the lightcurve fit is	1.05 %
Best Comparison Star	None
Optimal Aperture	3.97
Optimal Annulus	8.61
Transit Duration (day)	0.0563 +/- 0.0061

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.193 ± 0.021 vs 0.16309 (deviation 1.42σ)
Duration fit vs published	0.0563 d vs 0.0567 d (deviation 0.06σ)
Mid-time O–C	-0.4 min
Depth SNR	9.2
Residual scatter	1.05 %

Light curve

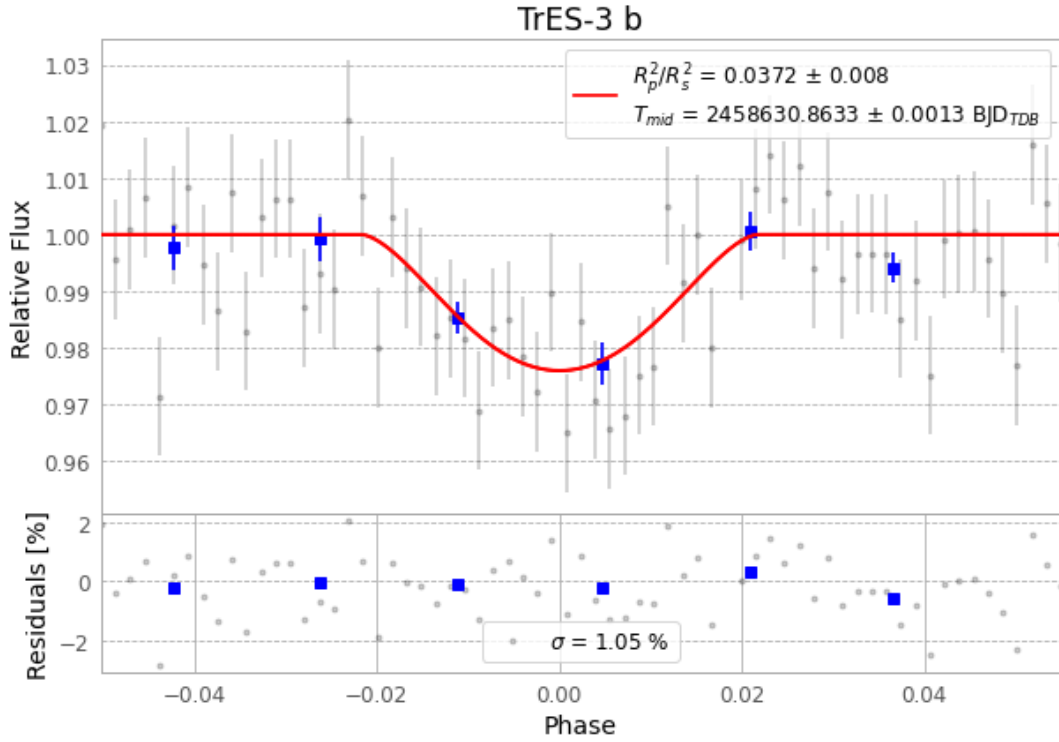


Figure 1 — FinalLightCurve_TrES-3 b_2019-05-27.png (SHA-256 f994369d55f9a4c3...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [371, 175], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 de507b06d7a0740c...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

67 FITS frames (44 MB), TRES-3190527070312.FITS → TRES-3190527102112.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-190527.log (5e7937e3cea5...), FinalLightCurve_TrES-3 b_2019-05-27.png (f994369d55f9...), FinalLightCurve_TrES-3 b_2019-05-27.csv (5b0afc77c2a1...)

Compute resources

Wall time 135.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-01 07:52Z.